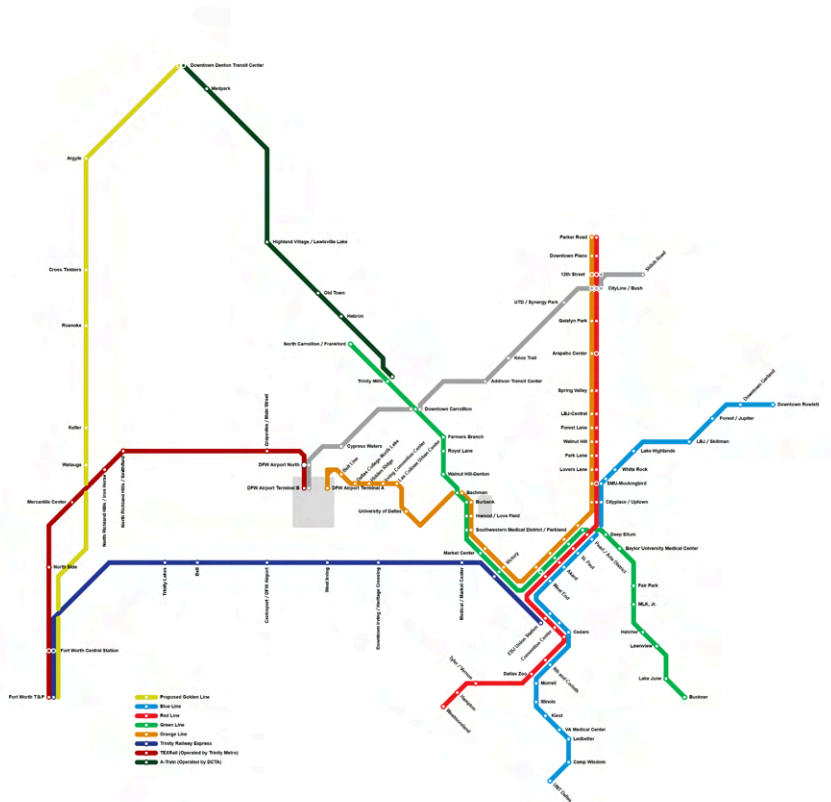


Executive Summary

One of the greatest struggles in the growing North Texas region is transportation. Our sprawling metropolis is connected by dozens of highways allowing Texans to go where they need to go. While our motorway system is great, connecting Texans with public transit is imperative to the growth of our major hub. With the completion and success of the Silver Line, the next step in expanding our public transit system is making the final edge of the Golden Triangle of Dallas, Fort Worth, and Denton. A new passenger train line connecting Fort Worth and Denton, and multiple cities in between will suit the needs of our growing metroplex.



Above is a passenger rail map of North Texas with the proposed Golden Line and its stations shown in Yellow.

Understanding the Needs of North Texas

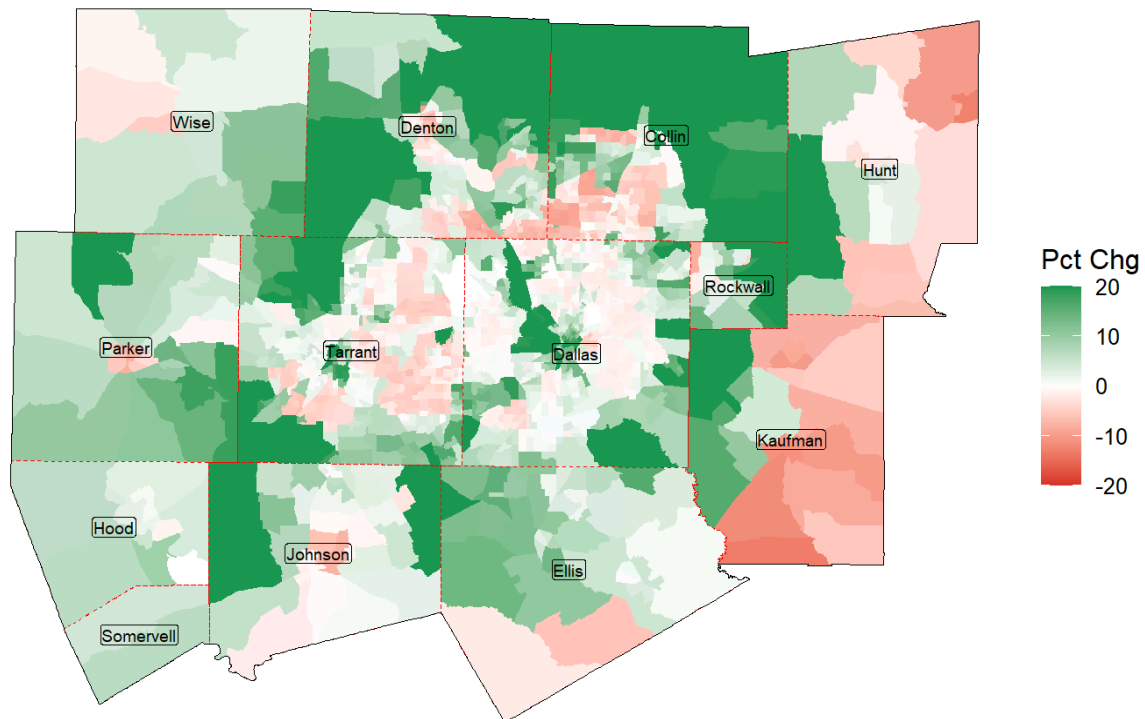
A transit desert is created when the supply does not meet the demand for transit in a given area. In a 2017 Study (Jiao), it was found that approximately 22% of the Fort Worth population is transit dependent. This is mostly comprised of elderly people, disabled people, and those who are otherwise unable to drive a vehicle. Transit is essential for these people and expanding their horizons in terms of travelling independently. This number will likely be affected in the future as economic hardship and rising price of gasoline impact those who drive. Currently, many of these transit dependent people do not have access to public transit in our metroplex. Much of these are in the suburbs, such as Roanoke, Haltom City, Watauga, Southlake, Argyle, all of which will be served by the Golden Line.

Those who commute by car also struggle to get around in rush-hour traffic around the metroplex. In a 2023 study (Parker), it was reported Fort Worth residents spend an average of 46 hours per year in road congestion. This number has increased over time, this means an increase in CO2 emissions, accident risk, and efficiency.

Denton and Tarrant county are growing rapidly in population. According to the 2010 and 2020 census, Denton county grew by 37%, and Tarrant county by 17%. That is approximately 500,000 new people over the course of 10 years. This rapid growth warrants rapid adaptation for the transit needs of the residents of this area.

Population 5 Year Forecast

Census Tracts of STI Market: Dallas-Fort Worth-Arlington, TX



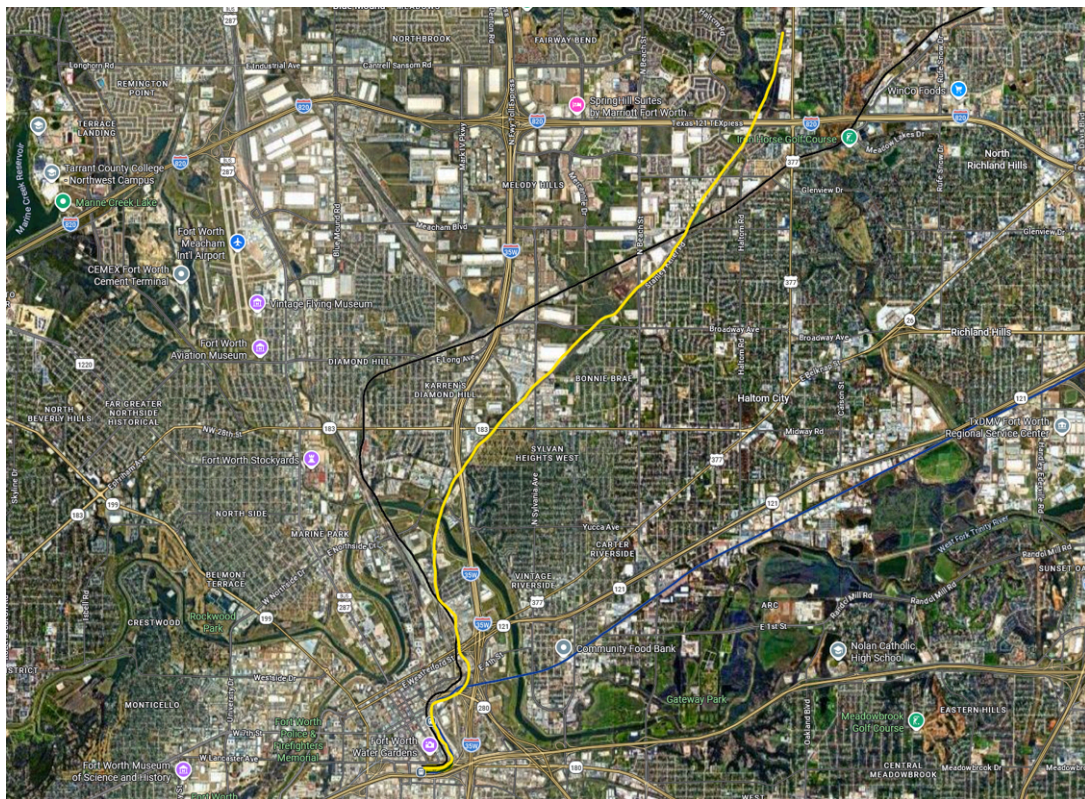
This map (Synergos) shows the forecasted population change by 2030. The Green areas in the south of Denton and North of Tarrant are the areas that will be serviced by the Golden Line. Rail stations are a great way to promote urban development in an area. Allowing more dense growth when less space is needed to accommodate vehicles. A great example of this is the Central Station in Denver, Colorado.



These images, provided by the US Geological Survey (2014-2023) show the development of the economic area of housing, shops and tourist attractions built around the train station after its construction. An expanded train station with new connections in Fort Worth and Denton, along with new transit opportunities in the cities along the way will pave the way for further development and housing opportunities for these growing cities (Brown, Huspeni, Proctor).

Full Project Description

The Golden Line will begin using the same tracks as the existing Trinity Railway Express and TEXRail services between Fort Worth T&P Station and Fort Worth Central Station. Then branch off using the Northbound tracks currently owned by UP. The two Fort Worth Stations need no further expansion as both are large enough to accommodate the new service. (Shown in Yellow)

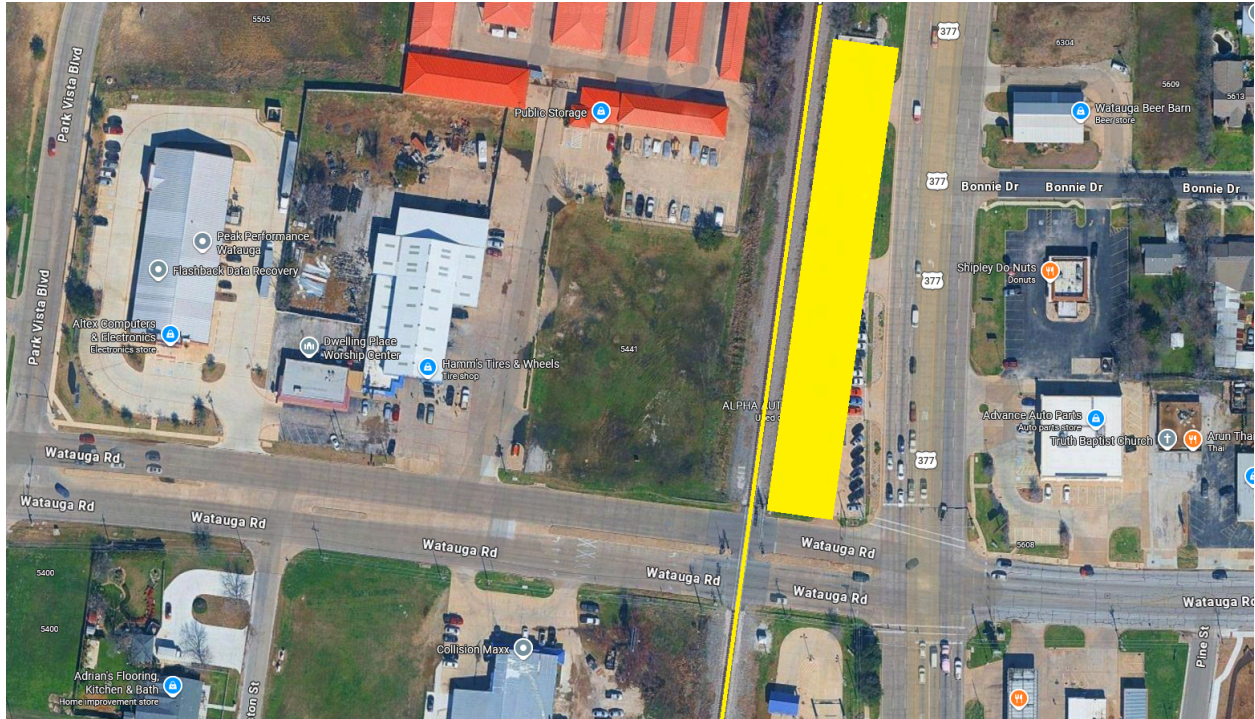


The line will continue on these tracks following US-377 north approximately thirty-six miles to Downtown Denton Transit Center where the existing tracks already run parallel to the terminus station of the DCTA A-Train service.



The yellow box shows the approximate area of construction of the expansion to the DDTC Station, along with the terminus of the Golden Line. The Downtown Denton and Fort Worth Stations are the most important and will generate the most economic development and fit right into the existing transit services of both cities.

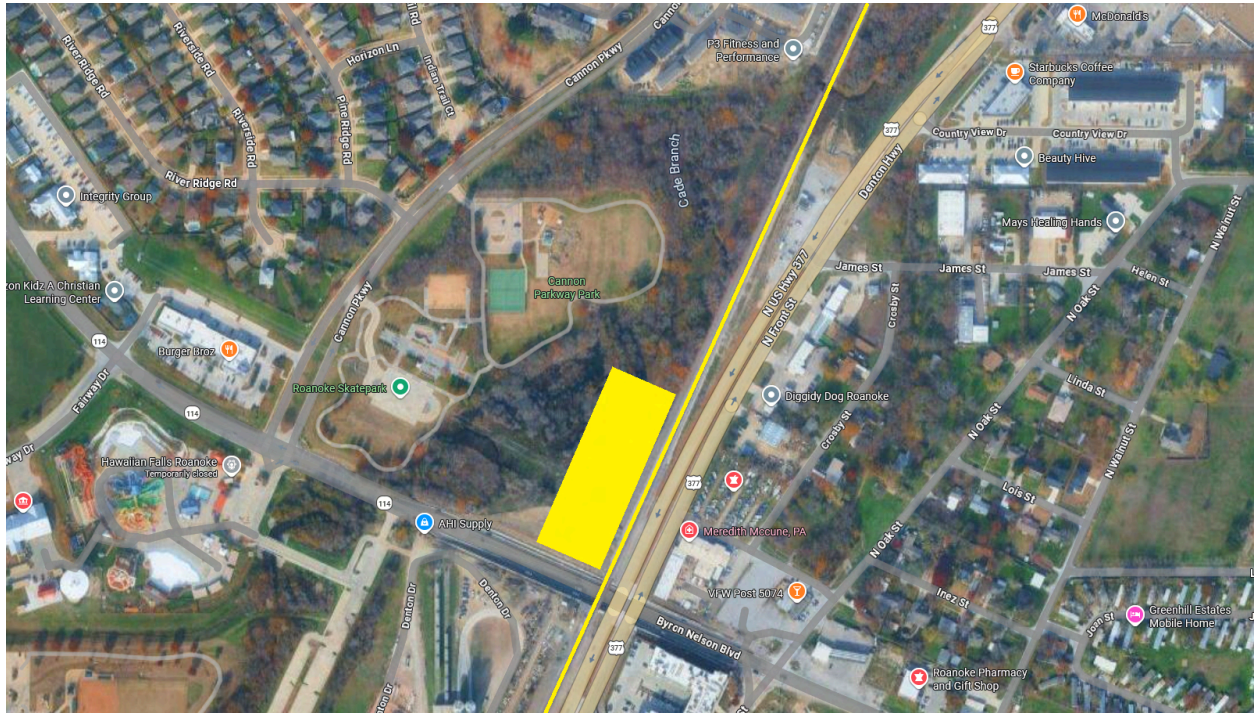
The first new station will be in Downtown Watauga on the corner of Watauga Road and US-377, providing easy access to local services in the downtown area, and easy connections for future bus routes to be added. This space is currently Alpha Autoplex.



The Keller Station will be on the corner of Golden Triangle boulevard and Katy Road. Servicing the downtown Keller area. This space is currently an empty plot of land owned by the Union Pacific Railroad.



The Roanoke station will service the downtown area of Roanoke, multiple residential neighborhoods, Hawaiian Falls, and multiple city parks. This land is currently owned by the City of Roanoke.



The Cross Timbers station will be at the corner of Cross Timbers Road and US-377. This station is in a relatively undeveloped area, and will serve as a park and ride station for surrounding cities of Argyle, Double Oak, Bartonville, Corral City, Flower Mound, and Lewisville. This station creates the perfect opportunity for future urban development of the area, and the opportunity for a future bus service connecting this station to the existing Old Town station in Lewisville. The land is currently privately owned.



The Argyle station will be located on the corner of Crawford Road and US-377. This station will service the surrounding residential areas of Argyle as well as the downtown area, multiple schools, and city parks. This land is currently owned by the city of Argyle.



(Satellite images and maps provided by Google)

In addition to the construction of these stations, the existing single main line will be expanded to a double main line, allowing rail traffic flowing in both north and south directions, with the existing sidings remaining as three-rail sections for passing between the freight and passenger traffic.

Using the standard set by the DART Silver Line, the Golden Line will start with Eight Stadler FLIRT Light Rail rolling stock cars. These are Electric-Diesel vehicles used around the world for rail services, with a top speed of 80-120mph. Speeds and timings of the Golden Line service will be determined at a later date with discussions between city agencies, Union Pacific Railroad, and real-world tests once the infrastructure is completed. (Stadler)



Stadler FLIRT operated by Keolis in Amersfoort, Netherlands. (Image: Roel Hemkes)

Expected Outcomes

The Golden line is expected to greatly improve many qualities of life in the regions it serves and Dallas-Fort Worth as a whole. This includes reduction in travel times on road ways, overall pollution, reduction in car dependancy, accessibility, and generation of revenue in the long term.

In the short term, before the completion of the line, this project will generate jobs and opportunities for construction workers, planners, and other city officials. Once the train line is completed, the next year or so will see a slight economic increase in the areas around the stations, as well as an increase in revenue from fares, and a slight reduction in congestion on roadways. The next few years after the opening, this service is expected to be integrated in the lives of residents of all serviced areas. This will allow transit reliant individuals much more freedom to travel in North Texas. This line will be a cheaper, more reliable alternative to driving, and with proper integration and support from cities, it will be faster, and more convenient than driving as well.

Long-term success of the Golden Line will have numerous outcomes for the region. This means less emissions, air quality improvement, and less fuel consumption. Economic development will also increase, with less space needed for vehicles, surrounding areas can develop denser, and more efficiently. This has been shown to have real impact by many studies, including one analyzing United States light rail systems, including previous projects in Dallas-Fort Worth (Arndt). Success of the Golden Line will pave the way for even more transit projects in the region, such as bus lines and further expansion of the rail system.

The revenue generated by the Golden Line in terms of fares will not directly offset the cost of construction and operation. The expected fare cost will be the same regional fare used by Trinity Metro, and Denton County Transit Authority. This is currently \$2.00 one-way and \$4.00 for a day pass. The long-term operation cost, and revenue generated will be largely from nearby property taxes and sales tax. These numbers are set to increase in any place near transit service, as found with past projects, according to a previous transit study, and by reports from DART and DCTA. This will be a significant return on investment in the long term for the federal and state grants needed to fund this project. (Latest Costs, Arndt)

Budget/Expenses

Land Acquisition:

(Note: These prices are volatile and based upon estimations from Zillow, Tarrant Appraisal District, and Denton County Appraisal District)

Watauga - \$415,000 West side lot, \$85,000 East Lot.

Keller - \$217,000

Roanoke - \$160,000

Cross Timbers - \$3,300,000 (Note: This is the market value of the full 75-acre lot, the planned station will need approximately 7-10 acres, thus the entire plot of land would not need to be purchased. For the full budget purpose, a figure of \$500,000 will be used. An exact amount will be determined with the land owner at a later date.)

Argyle - \$575,000

Denton - \$580,000

Total: \$2,532,000

Requested Budget (Adjusted for future price increases, complications, etc.): \$3,000,000

Corridor Usage:

The Golden Line will be renting the corridor from Union Pacific. The cost of this is difficult to determine as this figure is often kept private between the railroad and contractor. This figure is also impacted by the construction needed to expand the corridor and how that will affect Union Pacific's ability to maintain their services during construction. An estimation cannot be made at this point in time.

Construction:

Rail - This will get a more concrete number at a later date. Using the data from DFW's most recent rail project, the Silver Line, an estimation can be made. The Silver Line's construction cost a total of approximately 2.1 billion dollars. This was a 26 mile project with 10 stations. The Golden line will be 36 miles, with 5 new stations, 1 expanded station, and 2 existing stations with minor accommodations needed. The Silver Line project also called for multiple new interchanges with other rail systems, as an existing rail corridor did not already exist in some of the length of the project. The Golden Line will be using existing corridors, though expanding it. The cost of this is difficult to estimate, as it will be a joint project with Union Pacific Railroad, who owns the corridor. The Silver Line was contracted to Jacobs Engineering for station architecture design and planning. Construction was contracted to The Walsh Group, who would handle construction equipment, labor, materials, etc. Many of the specifications of this and similar contracts are kept private. With more track distance, but less stations and complex track expansion needed, a comparable budget of \$2,000,000,000 will be used for overall construction.

This figure, however, is subject to change depending on future market changes, and deals with contracting companies.

Rolling Stock:

The FLIRT vehicles the Golden Line will purchase from Stadler vary in cost depending on various factors, and will be discussed with Stadler in the future. The expected cost of the Eight vehicles is \$15,000,000 per vehicle, totalling \$120,000,000. This figure is based on averages of recent contracts for the FLIRT vehicles published in reports on Stadler's website across various countries such as the United States, Poland, The Netherlands, Finland, and Switzerland. (Stadler)

Future Costs:

Operating costs for maintenance, wages of approximately 150 employees, and general operating expenses are expected to be similar to the DCTA A-Train, which operates on a similar scale in Denton County. The DCTA Annual expense for 2025 was \$63,000,000. The scale of the golden line will be slightly bigger with the longer track length and 2 more stations. A proportional budget of \$100,000,000 will be expected for annual operating costs. (DCTA)

Overall Budget:

The requested budget for initial investment of land acquisition, construction, and rolling stock, is \$2,103,000,000. This is not including future costs of operation and corridor usage, which will be determined at a later date. This is the initial investment to be requested in the form of State and Federal funding.

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